PERMANENT MAGNET SYNCHRONOUS MOTOR

BY GROUP 02

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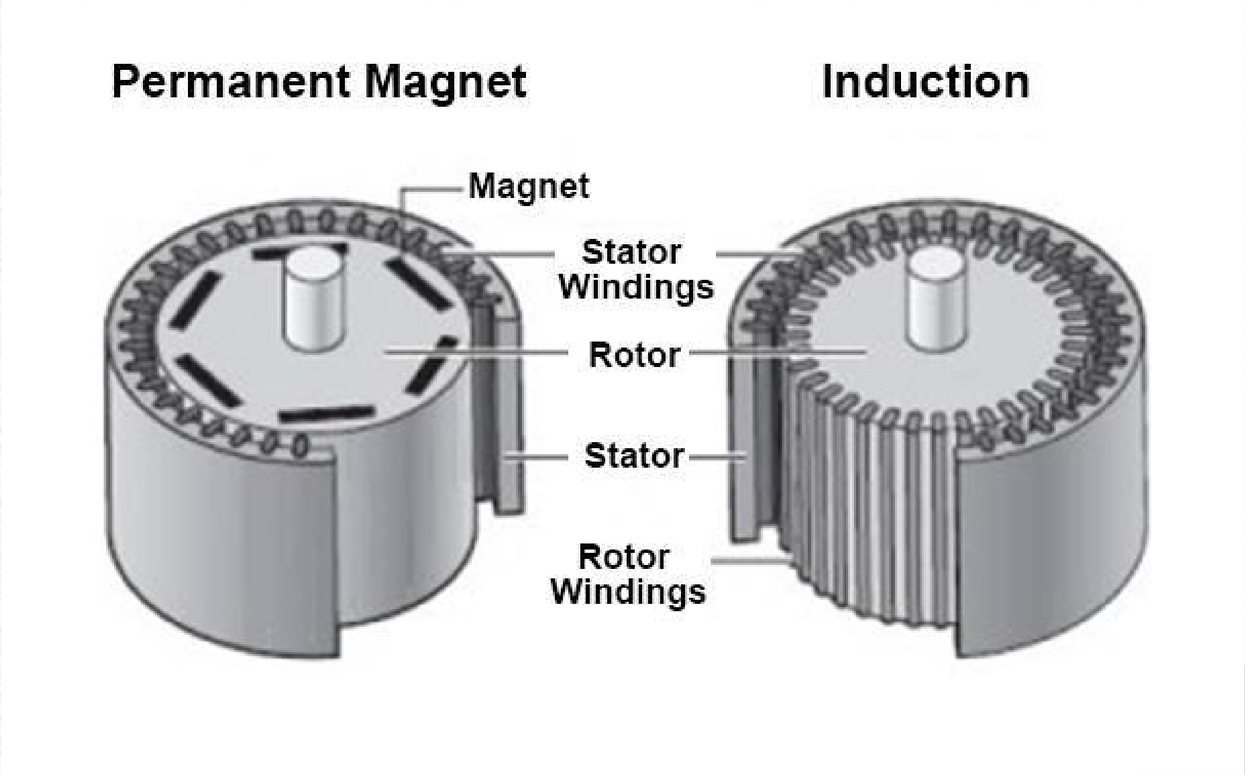
INTRODUCTION

What is PMSM?

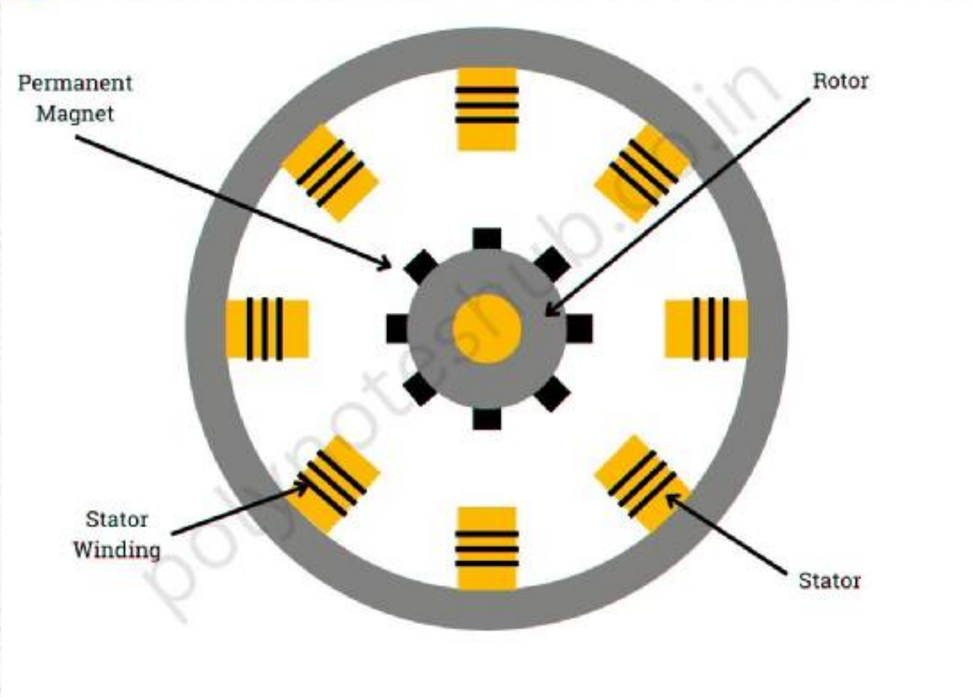
- The defining feature of Permanent Magnet Synchronous Motors is that the rotor's magnetic field is provided by permanent magnets, rather than by rotor current or field windings.

- Unlike induction motors, PMSMs operate synchronously, meaning the rotor rotates at the same speed as the stator's rotating magnetic field.



INTRODUCTION

Contruction of PMSM



1. **STATOR :**
- Core
- Windings

2. **ROTOR :**
- Contains Permanent Magnets instead of rotor windings
- Magnets like Neodymium, Samarium cobalt

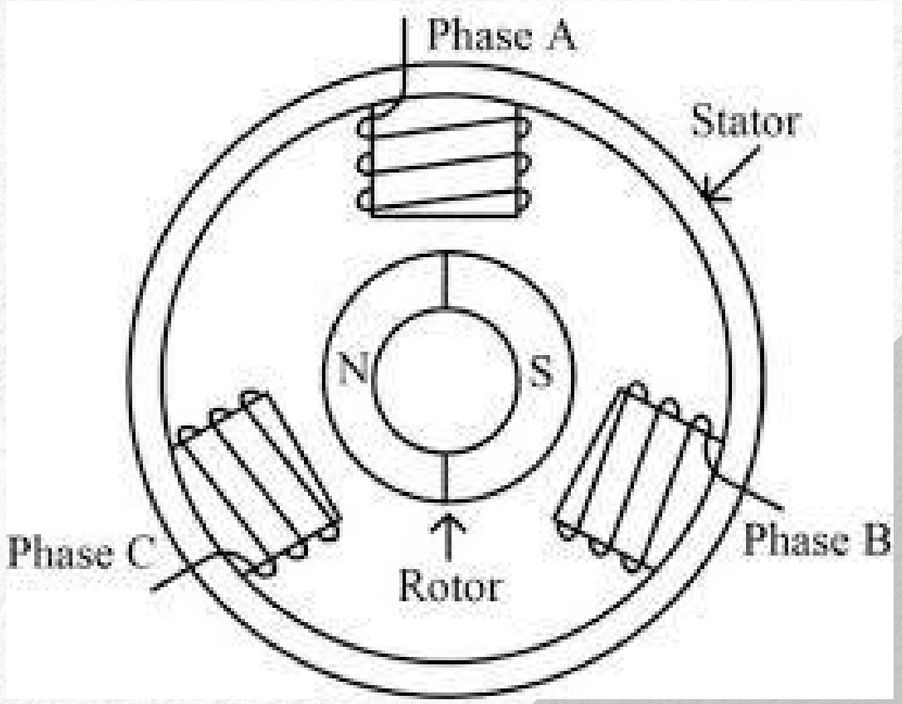
The rotor is what sets PMSMs apart. Instead of rotor windings or a squirrelcage structure, the rotor contains permanent magnets that produce a constant magnetic field without electrical input.

WORKING PRINCIPLE

Stator and Rotor field interaction

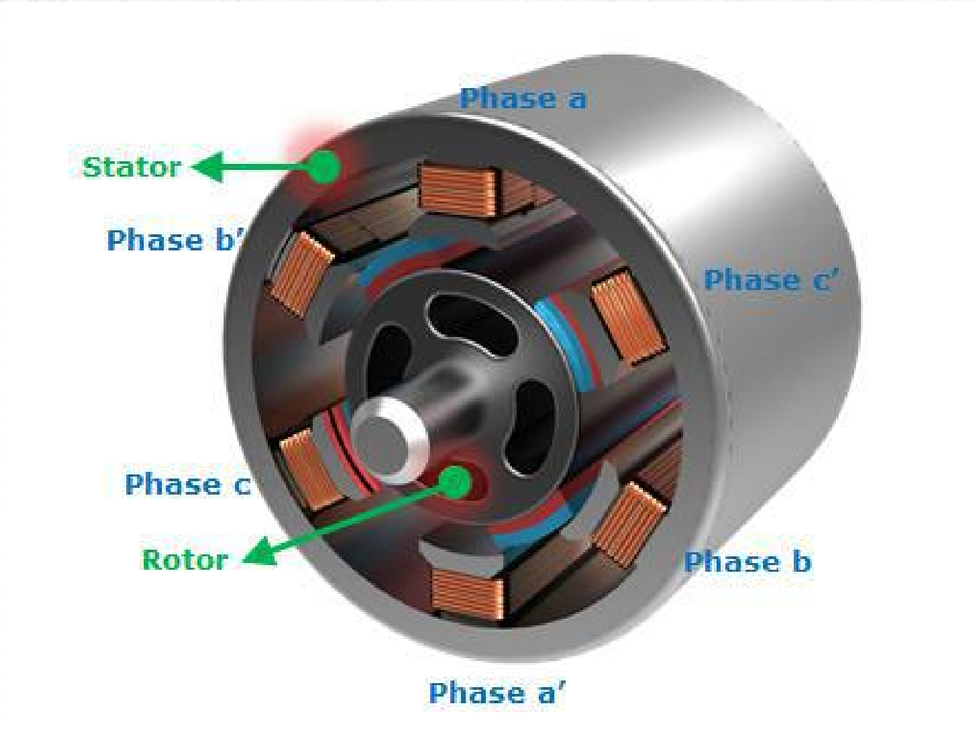
- The core operating principle of Permanent Magnet Synchronous Motors lies in the interaction between the stator's rotating magnetic field and the rotor's permanent-magnet field.

- When a threephase current energizes the stator, it creates a rotating magnetic field.



WORKING PRINCIPLE

Torque Generation



- The rotor magnets align with this field, and once synchronized, the rotor rotates at the same speed as the stator field.
- Torque is generated by the angular separation, known as the torque angle, between the stator field and the rotor field.
- Maintaining this torque angle within a controlled range is essential for smooth operation, high efficiency, and stable performance.

CONTROL METHOD

V/F control of PMSM

Priniciple: V/f = Φ [constant]

Working:

- Ns is directly proportional to frequency
- So as the frequency changes voltage changes but here we have to keep v/f value contant
- By keep that constant

Frequency Increases - speed increases

Frequency decreases - speed decreases

Speed ∝ Supply frequency

CONTROL METHOD

What if V/f is not Constant?

Case01: Under Fluxing

V/f = Φ

- f ↑ Flux ↓
- Torque ↓ leads to fail to drive the given load

Case02: Over Fluxing

V/f = Φ

- f ↓Flux ↑ High current leads to overheating

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THANK YOU